# Software Test Plan (STP)

## for

## Student Attendance Management System (SAMS)

Version: 1.1

Date: 29-10-2026

Authors: Team 1

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# 1. Introduction

This document defines the test plan for the Student Attendance Management System (SAMS) version 1.0. It gives the objectives, scope, approach, environment, schedule, responsibilities and test cases for testing SAMS against the requirements in the SRS.

**Objectives:**

- Verify all functional requirements (REQ-1 to REQ-19)
- Validate performance and security requirements (NFR-1 to NFR-11)
- Make sure every SRS requirement is covered by atleast one test case

**References:**

- SAMS SRS v1.0
- SAMS SAD v1.0
- IEEE 829 Standard for Software Test Documentation

# 2. Test Items

| **Module (Django app)** | **What it does** |
|---|---|
| accounts | Login, logout, roles, user management, Excel upload |
| academics | Subjects, classes, enrolment, faculty assignment |
| attendance | Mark / update attendance, percentage, change log |
| reports | Student view, PDF / Excel reports, shortage list |
| leaves | Leave requests, approval, shortage email |

Build under test: SAMS v1.0 from the main branch of the team GitHub repository.

# 3. Features to be Tested

| **Req ID** | **Feature** |
|---|---|
| REQ-1, REQ-2 | Login and role based dashboard |
| REQ-3, REQ-4 | Add / edit / delete users, Excel upload |
| REQ-5 to REQ-7 | Subjects, classes, faculty assignment |
| REQ-8, REQ-9 | Mark attendance, no duplicate marking |

| | |
|---|---|
| REQ-10 | Update attendance within 3 days |
| REQ-11 | Automatic percentage calculation |
| REQ-12 | Student views own attendance |
| REQ-13, REQ-14 | Report generation, PDF / Excel download |
| REQ-15 | Shortage list |
| REQ-16 | Shortage email |
| REQ-17 to REQ-19 | Leave apply, approve / reject, leave not counted absent |
| NFR-1 to NFR-3 | Performance |
| NFR-6 to NFR-11 | Security |

## 4. Features Not to be Tested

- Internal working of Django, MySQL, reportlab and openpyxl (assumed tested by thier developers)
- Gmail delivery after the mail is handed to the SMTP server
- Load above 100 users (outside SRS scope)
- Other languages (only English supported)

## 5. Test Approach / Strategy

| **Level** | **What** | **Tool** |
|---|---|---|
| Unit testing | Models, views and helper functions of each app | Django TestCase |
| Integration testing | Flows across apps, e.g. mark attendance -> shortage email | Django test client |
| System testing | Full end to end flows in the browser | Manual (browser) |
| Acceptance testing | Demo scenarios with the faculty | Manual |

**Types of testing:**

- Functional testing of every REQ

- Boundary testing - 74.99% vs 75%, 3 day edit limit
- Negative testing - wrong inputs, unauthorised access
- Performance testing - page load and report time
- Regression testing - rerun all unit tests before every merge to main

**Entry criteria:**

- Code merged to main and running on the test machine
- Test data loaded

**Exit criteria:**

- All High priority test cases executed
- Atleast 95% test cases passed
- No open critical defects

### 5.1 Security Validation

Security testing is done against the security objectives SO-1 to SO-3 and requirements NFR-6 to NFR-11 in the SRS.

| **Security Check** | **How it is tested** | **Req** |
|---|---|---|
| Password hashing | Check user table in MySQL - password should not be plain text | NFR-6 |
| Role based access | Login as each role and try to open pages of other roles - should get 403 | NFR-7 |
| Student data privacy | Student changes URL to view another student's attendance - should get 403 | NFR-8 |
| Session timeout | Stay idle for 31 minutes - should be logged out | NFR-9 |
| CSRF protection | Submit a form without CSRF token - should be rejected | NFR-10 |
| SQL injection | Enter ' OR 1=1 -- in login and search fields - should not work | NFR-10 |
| Audit log | Edit an attendance record - log entry should be created with name and time | NFR-11 |
| Sensitive data in logs | Check log files - no passwords or phone numbers | SO-3 |

## 6. Test Environment

| Item | Details |
|---|---|
| Hardware | Team laptops (MacBook Air / Windows laptops, 8 GB RAM), Android phone for mobile view |
| OS | macOS 14, Windows 11 |
| Software | Python 3.10, Django 4.2, MySQL 8.0 |
| Browsers | Chrome, Firefox, Safari (latest) |
| Test data | 2 classes, 5 subjects, 70 students, 4 faculty, 1 admin |

## 7. Test Schedule

| Milestone | Date |
|---|---|
| Test plan and test case design | 05-10-2026 |
| Test environment setup | 12-10-2026 |
| Unit and integration testing (Sprint 1) | 19-10-2026 to 23-10-2026 |
| Unit and integration testing (Sprint 2) | 26-10-2026 to 30-10-2026 |
| System and security testing | 28-10-2026 to 30-10-2026 |
| Regression and acceptance testing | 30-10-2026 |
| Test summary report | 31-10-2026 |

## 8. Test Deliverables

- Test Plan (this document)
- Test cases
- Unit test scripts (tests.py in each app)
- Test data
- Defect list (JIRA bugs)
- Test summary report

## 9. Roles and Responsibilities

| Role | Name | Responsibility |
|---|---|---|
| QA Lead | Epari Subhransi | Prepare test plan, coordinate testing, report results, test reports module |
| Test Engineer | Cheethirala Sai Santosh | Test accounts module and security checks |
| Test Engineer | Deepak B | Test academics and attendance modules, performance tests |
| Test Engineer | Gyaan S Shetty | Test leaves module and shortage email, maintain test environment |

## 10. Risks and Mitigation

| Risk | Mitigation |
|---|---|
| Build not ready on time | Test each module on its own branch first |
| Test machine / MySQL not working | Each member has local MySQL setup |
| Email cannot be checked | Use Django console email backend while testing |
| Less time at end of semester | Write unit tests along with code in every sprint |

## 11. Assumptions & Dependencies

- MySQL and Django are installed on all team laptops
- Test data represents a real class (upto 70 students)
- A Gmail account is available for email testing
- Requirements in SRS v1.0 will not change during testing

## 12. Suspension & Resumption Criteria

**Suspend testing if:**

- Login or attendance marking is not working (blocks other tests)
- A critical security defect is found
- Test environment is down for more than 4 hours

**Resume when:**

- The blocking defect is fixed and retested
- Environment is back up

## 13. Test Case Management & Traceability

Test cases are given below. Each test case is mapped to SRS requirements, and the traceability table shows that every requirement is covered.

### Test Cases

| TC ID | Req ID | Test Description | Precondition | Expected Result | Priority | Result |
|---|---|---|---|---|---|---|
| TC-01 | REQ-1 | Login with valid ID and password | User exists | Dashboard opens | High | Pass |
| TC-02 | REQ-1 | Login with wrong password | User exists | "Invalid username or password" shown | High | Pass |
| TC-03 | REQ-2, NFR-7 | Student tries to open admin page | Logged in as student | 403 Forbidden shown | High | Pass |
| TC-04 | REQ-4 | Upload Excel with 10 students (2 with missing SRN) | Logged in as admin | 8 students added, 2 rows rejected with reason | Medium | Pass |
| TC-05 | REQ-5, REQ-6 | Add subject and class, assign students | Logged in as admin | Subject and class saved | High | Pass |
| TC-06 | REQ-8 | Mark attendance with 3 absentees for class of 60 | Faculty assigned to subject | 57 Present, 3 Absent saved | High | Pass |
| TC-07 | REQ-9 | Mark attendance again for same subject, date and hour | TC-06 done | Error "Attendance already marked" | High | Pass |
| TC-08 | REQ-10 | Edit attendance after 2 days and after 4 days | Record exists | 2 days - saved; 4 days - blocked | High | Pass |
| TC-09 | REQ-11 | Check % after 3 of 4 classes attended | Records exist | 75.00% shown | High | Pass |
| TC-10 | REQ-12, NFR-8 | Student opens own attendance; then changes URL to | Logged in as student | Own data shown; other student - 403 | High | Pass |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | another student | | | | |
| TC-11 | REQ-13, REQ-14 | Generate report for a date range in PDF and Excel | Attendance data present | Both files download with correct totals | High | Partly (export pending) |
| TC-12 | REQ-15 | Shortage list with students at 74.99%, 75%, 80% | Threshold 75% | Only 74.99% student listed | High | Pass |
| TC-13 | REQ-16 | Student attendance drops below 75% | Student email set | Shortage email sent to student | Medium | Pass |
| TC-14 | REQ-17 to REQ-19 | Apply leave, faculty approves | Student absent on that day | Status Approved; day marked Leave, % updated | Medium | Pass |
| TC-15 | NFR-2 | Submit attendance for 70 students | Test data loaded | Saved in less than 2 seconds | Medium | Not run |
| TC-16 | NFR-6 | Check password stored in database | User created | Hashed value (pbkdf2_sha256$... ) | High | Pass |
| TC-17 | NFR-9 | Stay idle for 31 minutes | Logged in | Session expired, redirected to login | Medium | Pass |

### Traceability Matrix

| **Requirement** | **Test Case(s)** |
|---|---|
| REQ-1 | TC-01, TC-02 |
| REQ-2 | TC-03 |
| REQ-3 | TC-05 (unit test) |
| REQ-4 | TC-04 |
| REQ-5 to REQ-7 | TC-05 |
| REQ-8 | TC-06 |
| REQ-9 | TC-07 |
| REQ-10 | TC-08 |
| REQ-11 | TC-09 |

| | |
|---|---|
| REQ-12 | TC-10 |
| REQ-13, REQ-14 | TC-11 |
| REQ-15 | TC-12 |
| REQ-16 | TC-13 |
| REQ-17 to REQ-19 | TC-14 |
| NFR-2 | TC-15 |
| NFR-6 | TC-16 |
| NFR-7 | TC-03 |
| NFR-8 | TC-10 |
| NFR-9 | TC-17 |
| NFR-10, NFR-11 | Section 5.1 checks |

## 14. Test Metrics & Reporting

- % of test cases executed
- % passed / failed
- Number of defects per module
- Defects by severity
- Requirement coverage %

Test status is reported at the end of every sprint and a final test summary report is made at the end.

Status as on 29-10-2026: 73 automated Django tests run in GitHub Actions on every pull request, all passing. TC-11 export part is pending (SEM-16) and TC-15 performance test is not run yet.

## 15. Approvals

| **Role** | **Name** | **Signature / Date** |
|---|---|---|
| QA Lead | Epari Subhransi | |
| Dev Lead | Deepak B | |

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|---|---|---|
| Team Lead | Gyaan S Shetty | |
| Faculty Guide | | |